

CPE Vegetable and Fruit Intake Frequency and Mortality in Patients With and Without Chronic Kidney Disease: A Hospital-Based Cohort Study



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Objective: Patients with advanced chronic kidney disease (CKD) are generally discouraged from consuming high amounts of vegetables and fruits given the potential risk of hyperkalemia. In the general population, however, lower vegetable and fruit intake is associated with higher mortality. This Japanese hospital-based prospective cohort study including non-CKD and CKD participants examined whether the frequency of vegetable and fruit intake is associated with mortality, and whether the presence of CKD modifies this association.

Methods: Participants were 2,006 patients who visited the outpatient department of a general hospital between June 2008 and December 2016 (55% men; mean age, 69 years). Among these participants, 902 (45%) and 131 (7%) were non-dialysis-dependent patients with CKD and hemodialysis patients, respectively. The frequency of vegetable and fruit intake was determined by a self-report questionnaire using an ordinal scale, “never or rarely,” “sometimes,” and “every day.” Multivariable Cox proportional hazards analysis was conducted, adjusting for potential confounders.

Results: Vegetable and fruit intake frequency decreased with worsening CKD stage (P for trend < .001). Baseline serum potassium levels stratified by CKD stage were similar across the three vegetable and fruit intake frequency groups. During a median follow-up of 5.7 years, 561 participants died (47.1/1,000 person-years). Adjusted hazard ratios relative to the “every day” group were 1.25 (95% confidence interval, 1.04-1.52) and 1.60 (95% confidence interval, 1.23-2.08) for the “sometimes” and “never or rarely” groups, respectively, after adjusting for demographic factors, comorbidities, and CKD status. When stratified by CKD status, a similar, albeit non-significant, dose-dependent relationship was observed between vegetable and fruit intake frequency and all-cause mortality irrespective of CKD status, with no effect modification by CKD status ($P_{interaction} = .69$).

Conclusion: A lower frequency of vegetable and fruit intake is significantly associated with a higher risk of death regardless of CKD status.

Key words: CKD; epidemiology; hemodialysis; hyperkalemia; nutrition

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Introduction

PATIENTS WITH ADVANCED chronic kidney disease (CKD), especially those on hemodialysis (HD), are generally discouraged from consuming high amounts

of vegetables and fruits given the potential risk of hyperkalemia. In the general population, however, higher vegetable intake and fruit intake are associated with lower mortality. Systematic reviews and meta-analyses of cohort studies

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have reported that a higher intake of vegetables and fruits is associated with a reduced risk of cardiovascular disease, cancer, and all-cause mortality.¹⁻³

Emerging evidence suggests that higher vegetable and fruit intake may be associated with lower mortality among adults with CKD, including those who are non-dialysis-dependent and those who are on HD.^{4,5} A systematic review of observational studies reported that dietary patterns high in fruit and vegetable intake are associated with 30% lower all-cause mortality among patients with mild to moderate CKD.⁴ A multinational prospective cohort study of adults treated with HD (the DIET-HD study) reported that an increase in vegetable and fruit intake to approximately 17 servings per week (2-3 servings/day) is associated with a 20% lower risk of all-cause mortality compared with the intake of about two servings per week.⁵ These studies, however, were conducted in non-Asian countries. Amounts of vegetable and fruit intake vary markedly across countries, and diets consumed by Asian populations are estimated to be higher in vegetable and fruit content relative to diets consumed by other populations.⁶

In this study, we used data from a Japanese hospital-based prospective cohort study to examine the association between vegetable and fruit intake frequency and all-cause mortality in non-CKD and CKD populations, and whether this association is modified by the presence of CKD.

Methods

Study Population

This study was part of an ongoing hospital-based prospective cohort study. Details have been described elsewhere.⁷⁻¹² Briefly, Japanese residents aged 20 years or older who visit the outpatient department of a general hospital on an island are eligible for this registry, which was initiated in June 2008. The general hospital has approximately 20 departments, as well as a dialysis center, and is the only major hospital on the island. Upon registration, information about demographic characteristics, lifestyle, and medical history is collected. After enrollment, clinical research coordinators review the complete medical record including recorded risk factors and outcomes using a standardized data collection form. All participants provided written informed consent. The ethics committee of our university approved the study protocol.

For the present study, data of participants who registered from June 2008 to December 31, 2016 ($n = 3,074$) were obtained. Only participants with baseline serum creatinine and urinalysis measurements (except for dialysis patients), information about baseline vegetable and fruit intake, and at least one follow-up assessment were included in the analysis. Participants who received peritoneal dialysis at baseline were excluded.

CKD Status

Participants were divided into the following three groups: non-CKD, non-dialysis-dependent CKD, and HD patients. HD patients were defined as patients receiving maintenance HD at baseline. Non-CKD was defined as an estimated glomerular filtration rate (eGFR) ≥ 60 mL/minute/1.73 m² without proteinuria. Proteinuria was defined as a dipstick urinalysis score of 1+ or greater (equivalent to ≥ 30 mg/dL). eGFR was calculated using the eGFR formula for Japanese individuals.¹³

To assess cross-sectional associations between baseline serum potassium levels and CKD stages, non-dialysis-dependent CKD was further categorized by CKD stage¹⁴: stage 1-2 (eGFR ≥ 60 mL/minute/1.73 m² with proteinuria), G3a (eGFR 45-59), G3b (eGFR 30-44), and G4-5 (eGFR < 30).

Vegetable and Fruit Intake

The frequency of vegetable and fruit intake was assessed by the following two multiple-choice questions: "How many days do you usually eat vegetable-centered dishes?" and "How many days do you usually eat fresh fruit?" Responses to these questions were classified using an ordinal scale of "never or rarely," "sometimes," and "every day." Participants who consumed both vegetables and fruits every day were classified as the "every day" group, those who never or rarely consumed either vegetables or fruits were classified as the "never or rarely" group, and the remaining participants were classified as the "sometimes" group. Detailed descriptions of the questions and responses are provided in [Supplemental Appendix 1](#).

Outcome Measurement

The outcome was all-cause mortality. Deaths during the study period were obtained from medical records, the obituary column of the local newspaper, and information provided by relatives and friends. The date of informed consent was used as the index date. Participants were followed from the index date until death or the end of the study period (December 31, 2021), whichever came first.

Baseline Covariates

Height (cm) and weight (kg) were measured by trained clinical research coordinators. Body mass index (BMI) was defined as weight in kilograms divided by the square of height in meters. Age was calculated as of the index date. Information on sex, smoking habit (nonsmoker, past smoker, and current smoker), and disease history (diabetes, hypertension, stroke, and myocardial infarction) was obtained using a questionnaire. Hyperkalemia was defined as a serum potassium concentration ≥ 5.0 mmol/L.

Statistics

To assess cross-sectional associations between vegetable and fruit intake frequency and serum potassium levels at baseline, kernel density plots and box-plots stratified by

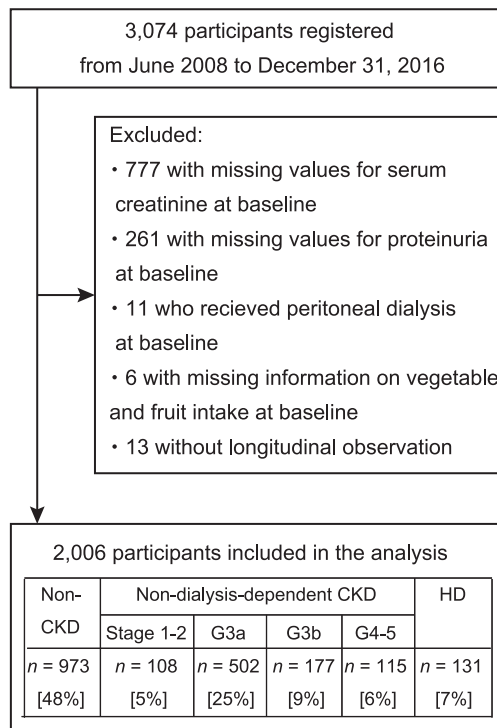


Figure 1. Flow diagram of participant selection. CKD, chronic kidney disease; HD, hemodialysis.

CKD stage were used. The Kaplan-Meier method was used to estimate the associations between vegetable and fruit intake frequency and mortality. We first examined the association between vegetable and fruit intake frequency and mortality in the entire study population to examine whether the association was independent after controlling for confounding factors and CKD status using Cox proportional hazards models. The assumption of proportional hazards was checked graphically and found not to be in violation. All models were analyzed unadjusted and adjusted for potential covariates identified *a priori*, including (i) sex and age (ii) demographic factors (i.e., sex, age, BMI, and smoking habit); as well as comorbidities (i.e., hypertension, diabetes mellitus, stroke, and myocardial infarction) (model 1); (iii) variables in model 1 and CKD status (model 2); and (iv) variables in model 2 and hyperkalemia (model 3).

The following analyses were performed to assess the robustness of the primary analysis. First, missing values were replaced by multiple imputation methods with 20 imputed datasets using all covariates and outcomes.^{15,16} Second, the analysis was repeated after adjusting for CKD stage instead of CKD status. Third, the analysis was repeated after excluding early deaths (<2 years) to rule out reverse causality. Fourth, the analysis was repeated by excluding HD patients and adjusting for eGFR category and proteinuria instead of CKD status. Fifth, subgroup analyses stratified by age, sex, diabetes, and hyperkalemia were

performed. Within each age stratum, age was adjusted as a continuous variable. To examine the effect modification by CKD of the association between vegetable and fruit intake frequency and mortality, stratified analysis by CKD status was conducted. The effect modification according to CKD status was assessed using the cross-product of CKD status and vegetable and fruit intake frequency. Finally, the effects of vegetable intake frequency and that of fruit intake frequency were examined separately. Analyses of vegetable intake frequency were adjusted for fruit intake frequency and vice versa.

All reported *P* values were 2-sided, and values < .05 were considered statistically significant. All statistical analyses were performed using IBM SPSS Statistics (version 27; SPSS, Chicago, Illinois) and Stata MP (version 16; Stata, College Station, Texas).

Results

Cohort Characteristics

A total of 2,006 participants aged ≥ 20 years, for whom data on baseline kidney function were available with no missing information on vegetable and fruit intake frequency, were included in this study (Fig. 1). Mean age was 69 years, 55% were men, 64% were hypertensive, and 39% were diabetic (Table 1). Participants who consumed both vegetables and fruits every day ("every day" group) accounted for 50%, and those who never or rarely consumed either vegetables or fruits ("never or rarely" group) accounted for 15%. When the frequency of vegetable intake and that of fruit intake were evaluated separately, 52% of participants consumed vegetables every day, and 91% of participants consumed fruits every day.

Non-dialysis-dependent CKD and HD patients accounted for 45% ($n = 902$) and 7% ($n = 131$) of the study population, respectively. Among the 131 HD patients, glomerulonephritis was the leading cause of renal failure (40%, $n = 53$), followed by diabetes (35%, $n = 46$), hypertension (5%, $n = 7$), and polycystic kidney disease (5%, $n = 6$). Median dialysis vintage was 4 years (interquartile range: 1–9 years).

Non-dialysis-dependent patients with CKD and HD patients who consumed both vegetables and fruits every day accounted for 53% and 28% of the study population, respectively. HD patients had a lower vegetable intake frequency than non-CKD participants. Fruit intake frequency was similar between the two CKD groups. Notably, 88% of HD patients consumed fruits every day.

Participant characteristics according to vegetable and fruit intake frequency are summarized in Table S1. Participants with a higher vegetable and fruit intake frequency were more likely to be older, be female, and have never smoked cigarettes, and were less likely to have CKD and a history of stroke. Baseline serum potassium levels were similar across all vegetable and fruit intake frequency groups.

Table 1. Baseline Characteristics of Participants by CKD Status (n = 2,006)

	No. of Missing Data (%)	Overall	Non-CKD (n = 973 [48%])	Non-Dialysis-Dependent CKD (n = 902 [45%])	Hemodialysis (n = 131 [7%])
Age, y	0	69 (11)	66 (11)	62 (10)	68 (11)
Male sex	0	1,102 (55%)	510 (52%)	510 (57%)	82 (63%)
Vegetable and fruit intake frequency	0				
Never or rarely		255 (15%)	105 (11%)	110 (12%)	40 (31%)
Sometimes		746 (37%)	374 (38%)	318 (35%)	54 (41%)
Every d		1,005 (50%)	494 (51%)	474 (53%)	37 (28%)
Vegetable intake frequency	0				
Never or rarely		247 (12%)	102 (10%)	106 (12%)	39 (30%)
Sometimes		715 (36%)	357 (37%)	305 (34%)	53 (40%)
Every d		1,044 (52%)	514 (53%)	491 (54%)	39 (30%)
Fruit intake frequency	0				
Never or rarely		27 (1%)	11 (1%)	13 (1%)	3 (2%)
Sometimes		163 (8%)	90 (9%)	60 (7%)	13 (10%)
Every d		1,816 (91%)	872 (90%)	829 (92%)	115 (88%)
BMI, kg/m ²	14 (0.7)	23.9 (4.0)	24.2 (4.1)	24.1 (3.8)	21.0 (3.1)
Smoking habit	1 (0.0)				
Current		264 (13%)	147 (15%)	97 (11%)	20 (15%)
Past		634 (32%)	271 (28%)	313 (35%)	50 (38%)
Never		1,107 (55%)	554 (57%)	492 (55%)	61 (47%)
Comorbidities					
Hypertension	4 (0.2)	1,286 (64%)	530 (55%)	655 (73%)	101 (77%)
Diabetes	0	775 (39%)	403 (41%)	323 (36%)	49 (37%)
Stroke	2 (0.1)	266 (13%)	93 (10%)	147 (16%)	26 (20%)
Myocardial infarction	2 (0.1)	95 (5%)	30 (3%)	53 (6%)	12 (9%)
eGFR, ml/min/1.73 m ² *	0	61.7 (18.7)	74.1 (11.0)	48.3 (15.8)	N/A
Proteinuria*	0	336 (18%)	0	336 (37%)	N/A
Laboratory data					
Serum potassium, mmol/L	24 (1.2)	4.4 (0.5)	4.3 (0.4)	4.5 (0.5)	4.8 (0.8)

BMI, body mass index; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; N/A, not available.

Continuous variables are presented as mean (standard deviation), and categorical variables are presented as number (percentage). Percentages do not add up to 100% in some cases due to rounding.

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*Among those not on dialysis therapy at baseline.

Associations Between Vegetable and Fruit Intake Frequency, Serum Potassium Levels, and CKD Stage at Baseline

Overall, mean baseline potassium level was 4.4 ± 0.5 mmol/L (Fig. 2A). Kernel density plots revealed close to normal distributions of baseline serum potassium

levels in the non-CKD, stage 1–2, G3, and G3b groups, whereas the data were spread out in the G4–5 and HD groups (Fig. 2B). Box plots stratified by CKD stage revealed similar distributions of baseline serum potassium levels across all vegetable and fruit intake frequency groups (Fig. 3A). Vegetable and fruit intake frequency decreased

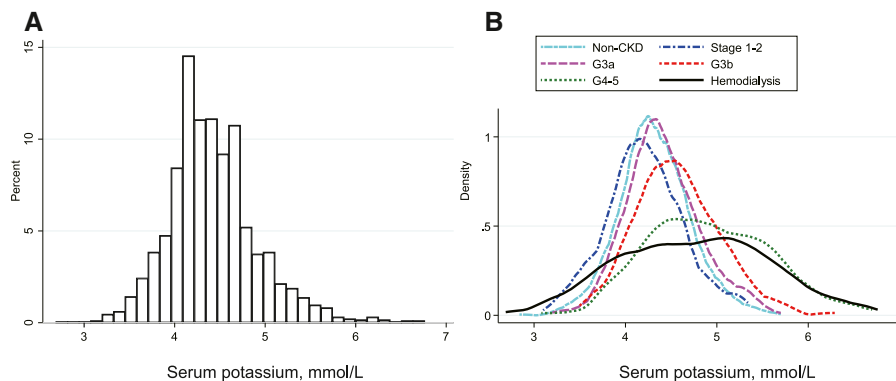


Figure 2. Distributions of serum potassium levels, overall (A) and by CKD stage (B). CKD, chronic kidney disease.

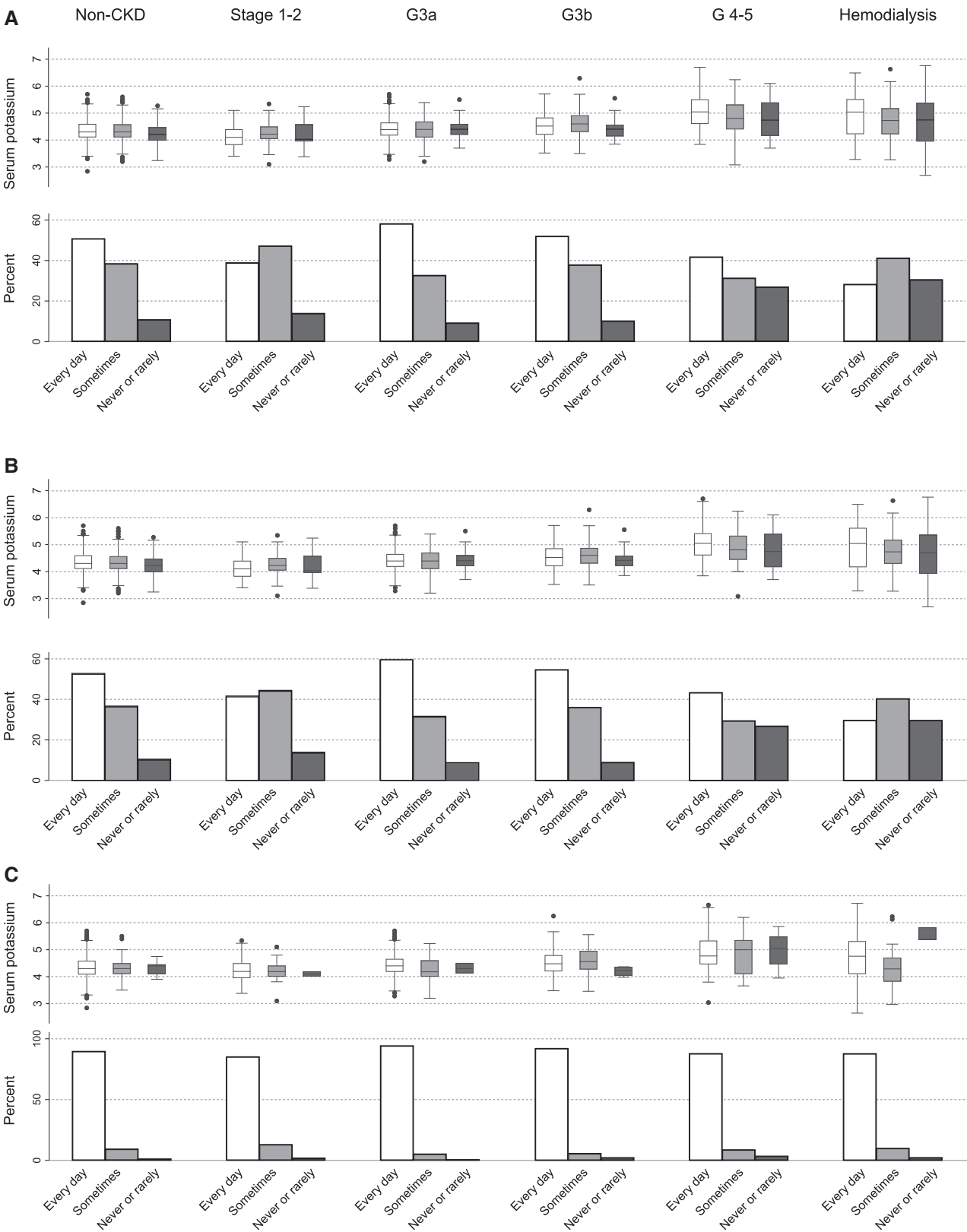


Figure 3. Box plots of serum potassium levels and percent frequency distributions by vegetable and fruit intake frequency (A), vegetable intake frequency (B), and fruit intake frequency (C), stratified by CKD stage. CKD, chronic kidney disease.

Table 2. Associations of Vegetable and Fruit Intake Frequency With All-Cause Mortality in the Entire Cohort

	Frequency of Vegetable and Fruit Intake			HR for trend†
	Never or rarely	Sometimes	Every d	
No. (%) of deaths	84 (33%)	197 (26%)	280 (28%)	
Mortality (95% CI)*	61.0 (49.2-75.5)	44.8 (39.0-51.5)	45.6 (40.5-51.2)	
HR (95% CI)				
Unadjusted	1.65 (1.29 - 2.11)	1.06 (0.88 - 1.27)	1 (Ref.)	1.22 (1.08 - 1.38)
Age/sex-adjusted	2.08 (1.62 - 2.66)	1.35 (1.12 - 1.63)	1 (Ref.)	1.42 (1.26 - 1.60)
Model 1	2.02 (1.57 - 2.60)	1.31 (1.08 - 1.58)	1 (Ref.)	1.39 (1.24 - 1.57)
Model 2	1.62 (1.25 - 2.10)	1.25 (1.03 - 1.52)	1 (Ref.)	1.27 (1.12 - 1.43)
Model 3	1.60 (1.23 - 2.08)	1.25 (1.04 - 1.52)	1 (Ref.)	1.26 (1.12 - 1.43)

CI, confidence interval; HR, hazard ratio.

Model 1: Adjusted for age, sex, body mass index, smoking habit, and comorbidities (i.e., diabetes, hypertension, stroke, and myocardial infarction).

Model 2: Model 1 + CKD status.

Model 3: Model 2 + hyperkalemia (≥ 5 mmol/L).

*per 1,000 person-years.

†HR for trend was calculated by entering the exposure categories as a continuous term in the Cox model.

with worsening CKD stage (P for trend $< .001$). When the frequency of vegetable intake and that of fruit intake were evaluated separately, box plots stratified by CKD stage showed similar distributions of baseline serum potassium levels between the vegetable (Fig. 3B) and fruit (Fig. 3C) intake frequency groups. The frequency of vegetable (P for trend $< .001$) but not fruit (P for trend = .71) intake significantly decreased with worsening CKD stage.

Association Between Vegetable and Fruit Intake Frequency and Mortality

During a median follow-up of 5.7 years (interquartile range: 5.4-7.3 years), 561 participants died (47.1/1,000 person-years). The Kaplan-Meier curves revealed that participants in the “never or rarely” group had increased cumulative all-cause mortality compared with those in the “every day” group (Fig. S1). After adjusting for age and sex, participants in the “never or rarely” group had a higher risk of mortality of 108% (hazard ratio = 2.08; 95% confidence interval, 1.62-2.66), whereas those in the “sometimes” group had a higher risk of mortality of 35% (hazard ratio = 1.35; 95% confidence interval, 1.12-1.63) compared with those in the “every day” group (Table 2). This association did not markedly change after further adjustment for BMI, smoking habit, and comorbidities (model 1). The association was partially attenuated but remained significant after further adjustment for CKD status (model 2) and hyperkalemia (model 3).

The results of the sensitivity analyses were consistent with those of the primary analyses. Imputing missing data (Table S2), adjusting for CKD stage instead of CKD status (Table S3), excluding early deaths (< 2 years) (Table S4), and excluding HD patients and adjusting for eGFR category and proteinuria instead of CKD status (Table S5) yielded similar results. Results were also similar when stratified by age ($P_{\text{interaction}} = .26$), sex ($P_{\text{interaction}} = .58$), diabetes ($P_{\text{interaction}} = .42$), and hyperkalemia ($P_{\text{interaction}} = .37$)

(Fig. 4). When stratified by CKD status, a similar, albeit non-significant, dose-dependent relationship was observed between vegetable and fruit intake frequency and all-cause mortality irrespective of CKD status, with no evidence of effect modification by CKD status ($P_{\text{interaction}} = .69$) (Table S6).

When the effects of vegetable intake frequency and fruit intake frequency were examined separately, a similar but slightly weaker association was observed between vegetable intake frequency and mortality compared to that observed in the main analysis (Table S7). The association between fruit intake frequency and mortality, however, was not significant after adjusting for vegetable intake frequency (model 4). Subgroup analyses revealed no significant interactions (Fig. S2). When stratified by CKD status, no evidence of effect modification by CKD status on the association between vegetable ($P_{\text{interaction}} = .77$) or fruit ($P_{\text{interaction}} = .47$) intake frequency and all-cause mortality was observed (Tables S8 and S9).

Discussion

This Japanese hospital-based prospective cohort study revealed that the frequency of vegetable and fruit intake decreased with worsening CKD stage. Baseline serum potassium levels stratified by CKD stage were similar across the vegetable and fruit intake frequency groups. Compared with participants in the “every day” group, both participants in the “sometimes” group and those in the “never or rarely” group had an increased mortality risk in a dose-dependent manner after adjusting for demographic factors, comorbidities, and CKD status. The association was similar regardless of CKD status, with no effect modification by CKD status. Our results suggest that eating both vegetables and fruits every day may not be associated with increased serum potassium levels, and that it may reduce rather than increase all-cause mortality in patients with

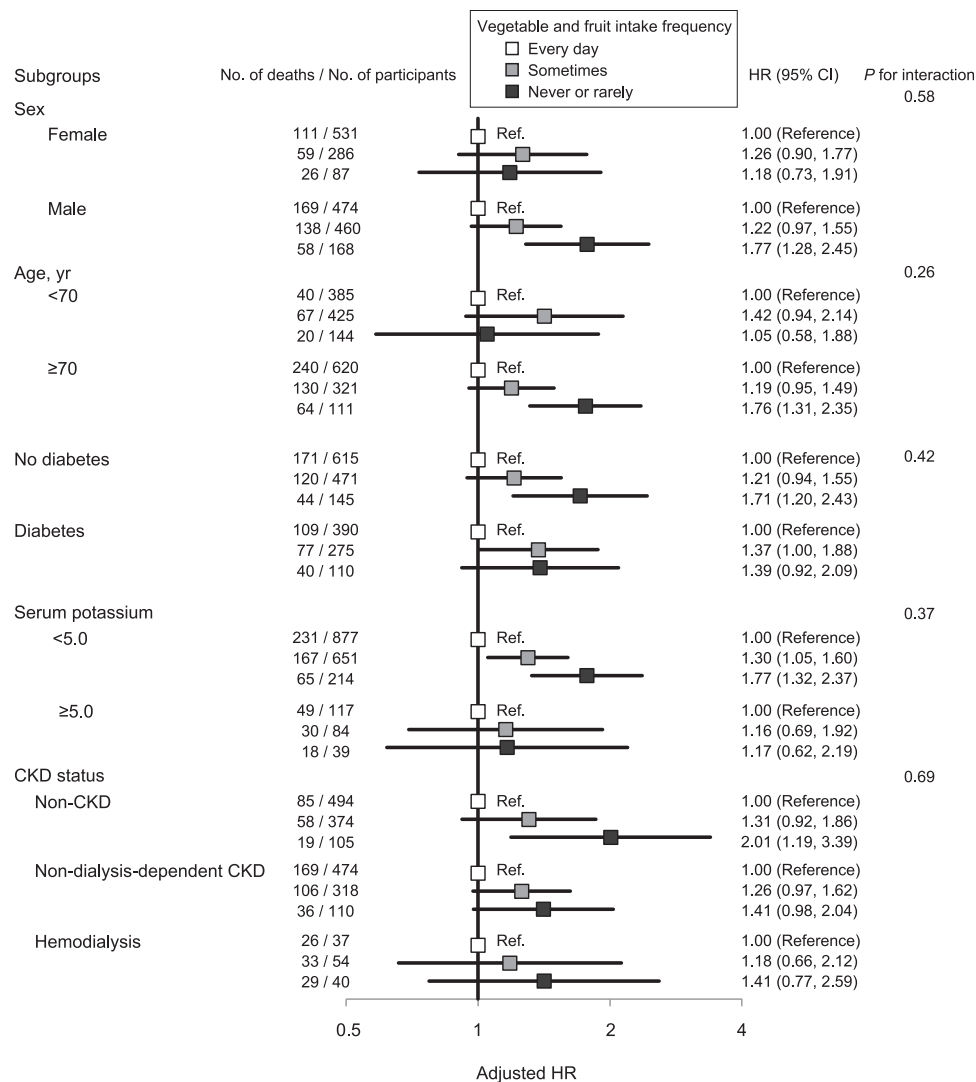


Figure 4. Subgroup analysis of the associations of vegetable and fruit intake frequency with all-cause mortality. Hazard ratios were adjusted for age, sex, BMI, smoking habit, comorbidities (i.e., diabetes, hypertension, stroke, and myocardial infarction), CKD status, and hyperkalemia. CI, confidence interval; HR, hazard ratio.

CKD including those on HD, as has been observed for non-CKD individuals.

Our baseline data showed that serum potassium levels stratified by CKD stage were similar across all vegetable and fruit intake frequency groups, suggesting that vegetable and fruit intake frequency is not associated with serum potassium levels even in patients with advanced CKD. Of course, there is a possibility of reverse causality, i.e., participants with hyperkalemia had a reduced frequency of vegetable and fruit intake because of their high potassium content. It is also possible that participants who consumed vegetables and fruits every day consumed them in reduced amounts or chose lower-potassium-containing vegetables and fruits. Moreover, it is likely that those who consumed vegetables and fruits every day soaked or boiled vegetables/fruits in water to remove potassium, which is a common dietary practice recommended for patients with

CKD in Japan. However, accumulating evidence suggests that dietary potassium levels are not associated with serum potassium levels or hyperkalemia in either non-dialysis-dependent-CKD or HD patients.^{17–20} Thus, vegetable and fruit intake frequency may be unrelated to serum potassium levels. Indeed, the present study showed that even patients with advanced CKD including those on HD consumed vegetables and fruits every day without increasing the risk of hyperkalemia. The lack of association between vegetable and fruit intake and serum potassium levels may be explained by the high fiber content of vegetables and fruits, their alkalizing potential, and reduced bioavailability of potassium.²⁰

In the present study, lower vegetable and fruit intake was associated with higher mortality among adults with CKD, including both non-dialysis-dependent CKD and HD patients, with no effect modification by CKD status. These

results suggest that similar associations reported in previous studies from non-Asian countries^{4,5} are also observed in Asian populations, whose diets are estimated to contain relatively high amounts of vegetables and fruits.⁶ Our findings add evidence supporting that, although food culture differs by country, the survival benefit of high vegetable and fruit intake may be universal.

Dietary fiber intake may be related to the association between vegetable and fruit intake and mortality. Vegetables and fruits are important sources of dietary fiber, and high dietary fiber intake has been associated with a lower risk of inflammation and mortality in individuals with kidney dysfunction.²⁰⁻²³ High fiber intake also has multiple health benefits, such as an increase in stool output, enhanced growth of beneficial microbiota, improved gut barrier function, decreased inflammation, and decreased uremic toxin production.²³ The alkalinizing effect of vegetables and fruits on metabolic acidosis may be associated with a reduced risk of mortality. Metabolic acidosis in CKD, a risk factor for death, improved comparably with fruit and vegetable intake and standard oral sodium bicarbonate treatment according to a post hoc analysis of a clinical trial.²⁴ Notably, fruit and vegetable intake improved measured health outcomes, such as systolic blood pressure, BMI, and low-density lipoprotein cholesterol,²⁴ suggesting that the beneficial effect of vegetable and fruit intake is not limited to its alkalinizing effect. Dietary magnesium may also explain the association between vegetable and fruit intake and mortality, because vegetables and fruits are also good sources of magnesium. Low serum magnesium levels have been shown to be associated with increased mortality in both non-dialysis-dependent CKD and HD patients.²³⁻³⁰ Magnesium intake is the most important determinant of serum magnesium concentration in HD patients.³¹ Other components of fruits and vegetables, such as vitamin C, carotenoids, and phytochemicals, may similarly contribute to a reduced risk of mortality.

The strengths of this study include its prospective design, adjustments for multiple potential confounders, and inclusion of patients with a wide range of kidney function (i.e., non-CKD, non-dialysis-dependent CKD, and HD patients). Moreover, as this was a single hospital-based study, there is no inter-hospital variation. The present study also has several limitations worth noting. First, vegetable and fruit intake frequency was self-reported and assessed only at baseline. It is possible that intake frequency changed over the course of the study period. In addition, we evaluated the frequency but not the quantity of vegetable and fruit intake. The fruit and vegetable intake questions have not been validated against any other more comprehensive tool. Moreover, adjustment for total energy intake was not possible because of limited data. Second, CKD was defined using one creatinine measurement and urinalysis at baseline, which could have led to a misclassification of some participants. Third, residual confounding effects of

unmeasured variables (e.g., socioeconomic status, residual kidney function, and potassium binder use) associated with vegetable and fruit intake frequency cannot be denied. Fourth, since this study was conducted in Japan, the results might not be generalizable to other populations, as dietary patterns widely vary across countries. Finally, due to the study design, causality between vegetable and fruit intake and mortality could not be established. Sensitivity analyses excluding early deaths (<2 years) to rule out reverse causality, however, showed similar results with the main analysis, suggesting that the possibility that patients with a preexisting higher risk of mortality might also be less inclined to eat healthier foods is low.

In conclusion, this Japanese hospital-based study revealed that the frequency of vegetable and fruit intake decreases with worsening CKD stage. Compared with participants who consumed vegetables and fruits every day, both participants who sometimes and never/rarely consumed vegetables and fruits had an increased mortality risk in a dose dependent manner after adjusting for demographic factors, comorbidities, and CKD status. This association was similarly observed regardless of CKD status, with no effect modification by CKD status, suggesting that consuming both vegetables and fruits every day may reduce all-cause mortality in patients with CKD as well as in non-CKD individuals.

Practical Application

This study provides evidence that a lower frequency of vegetable and fruit intake is significantly associated with a higher risk of death regardless of CKD status. Our findings suggest that daily intake of both vegetables and fruits may reduce all-cause mortality in patients with CKD including those on HD, as has been observed for non-CKD individuals.

CRedit authorship contribution statement

Minako Wakasugi: Conceptualization, Data curation, Formal analysis, Visualization, Writing – original draft, preparation. **Akio Yokoseki:** Conceptualization, Data curation, Project administration, Writing – review & editing. **Masakazu Wada:** Investigation, Writing – review & editing. **Takeshi Momotsu:** Conceptualization, Investigation, Writing – review & editing. **Kenji Sato:** Conceptualization, Investigation, Writing – review & editing. **Hiroyuki Kawashima:** Supervision, Writing – review & editing. **Kazutoshi Nakamura:** Supervision, Writing – review & editing. **Osamu Onodera:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Ichiei Narita:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

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Supplementary Data

Supplementary data related to this article can be found at <https://doi.org/10.1053/j.jrn.2023.01.011>.

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